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Würth Group

Hiring Team

Application: Master Thesis, Data and AI Intelligent Pricing Prototype for External Signal Integration

Dear Hiring Team,

I am applying for the Master Thesis on Intelligent Pricing at the Würth Group. The task of designing a prototype that integrates external market signals, macro indicators, and seasonal effects into a pricing framework lines up directly with the pipeline and modelling work I have already delivered.

In my Real Time Flight Tracking pipeline I ingested live positions every 30 seconds from the OpenSky Network API and joined each observation to airport, aircraft, and weather sources on Google Cloud with PySpark and dbt, orchestrated by Airflow with automated 15 minute refresh. That project shows I can identify credible external data sources, evaluate quality and freshness, and build robust features from them. In my Movie Analytics medallion pipeline on BigQuery I trained a BigQuery ML classifier with features deliberately split across two tables to prevent data leakage into pre release forecasts, exactly the discipline your brief mentions.

My CreditIQ credit scoring project shows how I document baselines, metrics, and trade offs in a regulator defensible way. I lifted the Disparate Impact ratio from 0.79 to 0.88 while cutting the false negative rate from 44 percent to 16.7 percent at 75 percent accuracy, and I wrote up the fairness accuracy trade off as a deliberate decision rather than an accident. My Economic Impact Analysis of Global Climate Events project applied Random Forest models to disaster duration and financial impact and communicated the findings to non technical stakeholders, which is close to the demo layer your brief mentions as an option.

I would welcome the chance to discuss the pricing prototype in more detail. Thank you for considering my application.

Kind regards,

Rahul Rawat